

Diabetes, Total Cholesterol, and Risk of Myocardial Infarction or Coronary Heart Disease Death: A Survival Analysis of the Framingham Heart Study

Michelle Navarrete Vega Chiamaka Lebechi

PM 518a: Applied Survival Analysis University of Southern California May 10, 2026

ABSTRACT

Background: Diabetes and elevated total cholesterol are modifiable risk factors for coronary heart disease. We examined their independent and joint associations with time to myocardial infarction or coronary heart disease (MI/CHD) death in the Framingham Heart Study. **Methods:** Among 4,354 participants free of CHD at baseline (1948), Kaplan-Meier curves and log-rank tests were used to compare survival by diabetes status. Cox proportional hazards models estimated univariate and multivariable associations; age was identified as a confounder and retained. Effect modification by diabetes on the cholesterol-MI/CHD association was evaluated via an interaction term. A secondary analysis categorized cholesterol into clinically defined groups. **Results:** Diabetic participants had substantially lower survival at all time points. In the multivariable model, diabetes (HR = 3.21, 95% CI: 2.35–4.39) and log-transformed total cholesterol (HR = 1.06 per 10-unit increase, 95% CI: 1.04–1.07) were independently associated with MI/CHD risk. No significant interaction between diabetes and cholesterol was detected ($P = 0.36$). **Conclusions:** Both diabetes and total cholesterol are significant, independent predictors of MI/CHD. Interventions targeting both risk factors simultaneously are warranted.

Keywords: survival analysis · Cox proportional hazards · diabetes · cholesterol · coronary heart disease · Framingham Heart Study

1. Data

The Framingham Heart Study is a long-term prospective cohort study initiated in 1948 to examine determinants of cardiovascular disease in residents of Framingham, Massachusetts. Cardiovascular outcomes were identified through hospital surveillance, participant reports, and death certificates. The primary outcome of interest was time to first nonfatal myocardial infarction or fatal coronary heart disease (MI/CHD) since the baseline examination.

The analysis was restricted to baseline measurements (period = 1). Participants with prevalent coronary heart disease at baseline were excluded from the dataset prior to receipt. Time to event was originally recorded in days and was converted to years by dividing by 365.25. Four participants with a survival time of zero were excluded, yielding a final analytic sample of $N = 4,354$ individuals.

2. Methods

2.1 Statistical Analysis

Baseline characteristics of study participants were summarized by diabetes status. Continuous variables are presented as mean (standard deviation) and categorical variables as frequency and percentage. Missing values are reported where applicable.

Kaplan-Meier survival curves were estimated separately for participants with and without diabetes to examine time to first

MI/CHD death. Median survival time and 95% confidence intervals were estimated for each group. Survival probabilities at 5, 10, 15, and 20 years were estimated with corresponding 95% CIs and compared between groups using the log-rank test. To account for multiple comparisons across five tests, a Bonferroni-corrected significance level of $\alpha^* = 0.05/5 = 0.01$ was applied.

Univariate associations of diabetes and total cholesterol with MI/CHD hazard were estimated using Cox proportional hazards models. Due to the wide variability in total cholesterol, a centered variable was created around the sample mean ($\bar{X} = 236.854$ mg/dL) and expressed per 10 mg/dL increase. Breslow's approximation was used to handle tied events. Martingale residuals with LOWESS smoothers were used to evaluate the linearity assumption for total cholesterol; three functional forms were assessed sequentially: centered, log-transformed, and log-transformed centered and scaled (per 10-unit increase). Results are provided in the Supplemental materials.

A multivariable Cox model was developed including diabetes and total cholesterol as primary exposures. Age, sex, and education were evaluated as potential confounders based on prior knowledge and their impact ($\geq 10\%$ change in the hazard ratio) on the primary exposure estimates; only age met this threshold and was retained (Table S1). The proportional hazards assumption was assessed using Schoenfeld residuals for the final model.

To assess effect modification, an interaction term between diabetes and log-transformed, centered, and scaled total cholesterol was added to the multivariable model. In a secondary analysis, total cholesterol was categorized into three clinically relevant groups: <200 mg/dL (healthy), 200–239 mg/dL (at risk), and ≥240 mg/dL (dangerous). Unadjusted and adjusted (for diabetes and age) Cox models were fit, with the healthy group as reference. Kaplan-Meier curves and a trend test were also conducted.

All analyses were conducted using Stata version 18.5. A two-sided significance level of $\alpha = 0.05$ was used unless otherwise specified.

3. Results

3.1 Baseline Characteristics

Baseline characteristics of the 4,354 participants by diabetes status are presented in Table 1. The majority did not have diabetes ($N = 4,240$, 97.4%); 114 participants (2.6%) had diabetes at baseline. Diabetic participants were older on average (55.2 vs. 49.7 years) and had higher total cholesterol (248.1 vs. 236.6 mg/dL). A higher proportion of diabetic participants were male (50.0% vs. 42.9%) and had some high school or less education (55.3% vs. 40.6%).

Table 1. Baseline characteristics of Framingham Heart Study participants by diabetes status, 1948 ($N = 4,354$).

Characteristic	No DM ($n = 4240$)	DM ($n = 114$)	Total
Age (years)	49.7 (8.6)	55.2 (7.6)	49.8 (8.6)
Total cholesterol ^a	236.6 (44.3)	248.1 (56.3)	236.9 (44.6)
Sex, n (%)			
Male	1821 (42.9)	57 (50.0)	1878 (43.1)
Female	2419 (57.1)	57 (50.0)	2476 (56.9)
Education, n (%) ^b			
Some HS or less	1722 (40.6)	63 (55.3)	1785 (41.0)
HS/GED	1244 (29.3)	24 (21.1)	1268 (29.1)
Some college	689 (16.2)	15 (13.2)	704 (16.2)
College grad+	477 (11.2)	9 (7.9)	486 (11.2)

Mean (SD) for continuous; N (%) for categorical.

^aMissing for 52 participants. ^bMissing for 111 participants.

3.2 Kaplan-Meier Analysis

The Kaplan-Meier survival curves by diabetes status are shown in Figure 1. Median survival time was not reached for non-diabetic participants within the follow-up period. Among diabetic participants, 50% experienced their first MI/CHD event by approximately 22.1 years (95% CI: 18.0, not estimable), reflecting a substantially higher event burden.

Survival was consistently lower among diabetic participants at all time points (Table 2), from 0.91 (95% CI: 0.83, 0.95) at

5 years to 0.57 (95% CI: 0.46, 0.67) at 20 years. Non-diabetic participants maintained markedly higher survival, from 0.98 at 5 years to 0.88 at 20 years. The absolute survival gap widened over time — approximately 7 percentage points at 5 years and 31 percentage points at 20 years. All four time-point log-rank tests were statistically significant after Bonferroni correction (all $\chi^2(1) > 97$, $P < 0.001 < \alpha^* = 0.01$). The overall log-rank test confirmed a significant difference across full follow-up ($\chi^2(1) = 97.29$, $P < 0.001$).

Table 2. Survival estimates and time-point tests by diabetes status.

Time	No DM S(t) (95% CI)	DM S(t) (95% CI)	$\chi^2(1)$	P
5 yr	0.98 (0.97, 0.98)	0.91 (0.83, 0.95)	98.06	<0.001
10 yr	0.95 (0.95, 0.96)	0.83 (0.74, 0.89)	100.64	<0.001
15 yr	0.91 (0.91, 0.92)	0.68 (0.58, 0.77)	103.98	<0.001
20 yr	0.88 (0.87, 0.89)	0.57 (0.46, 0.67)	123.37	<0.001

All $P < \alpha^* = 0.01$ (Bonferroni correction for 5 tests).

3.3 Cox Proportional Hazards Models

Univariate and multivariable results are presented in Table S2. In univariate analysis, diabetes was associated with a significantly higher risk of MI/CHD (HR = 4.14, 95% CI: 3.05–5.63, $P < 0.001$). Each 10 mg/dL increase in centered total cholesterol was associated with a 7% higher hazard (HR = 1.07, 95% CI: 1.06–1.09, $P < 0.001$).

Assessment of the linearity assumption using martingale residuals indicated that centered total cholesterol violated linearity (Figure S1, Panel A). A log transformation resolved the non-linearity (Figure S1, Panel B); this log-transformed, centered, and scaled form was retained in all subsequent models.

Age was identified as the only confounder meeting the $\geq 10\%$ threshold (18.1% change in diabetes HR; Table S1). The final multivariable model included diabetes, log-transformed total cholesterol, and age. Diabetes was associated with a 3.21-fold increased hazard (HR = 3.21, 95% CI: 2.35–4.39, $P < 0.001$); log-transformed cholesterol with a 6% increased hazard per 10-unit increase (HR = 1.06, 95% CI: 1.04–1.07, $P < 0.001$); and age with a 4% increased hazard per year (HR = 1.04, 95% CI: 1.03–1.06, $P < 0.001$).

The proportional hazards assumption was not violated for any covariate: diabetes ($\rho = 0.024$, $P = 0.54$), log-cholesterol ($\rho = -0.044$, $P = 0.28$), age ($\rho = -0.063$, $P = 0.09$); global test $\chi^2(3) = 5.16$, $P = 0.16$.

3.4 Effect Modification

The interaction term between diabetes and log-transformed total cholesterol was not statistically significant (HR = 1.03, 95% CI: 0.97–1.08, $P = 0.36$), providing no evidence that the cholesterol–MI/CHD association differed by diabetes status. A stratified model was therefore not necessary.

3.5 Secondary Analysis: Categorical Cholesterol

In unadjusted analyses, compared to healthy cholesterol (<200 mg/dL), individuals with at-risk levels (200–239 mg/dL) had HR = 1.75 (95% CI: 1.33–2.30) and those with dangerous levels ($\geq$ 240 mg/dL) had HR = 2.54 (95% CI: 1.96–3.28; overall $P < 0.001$). After adjustment for diabetes and age, associations were attenuated but remained significant: HR = 1.53 (95% CI: 1.16–2.01) and HR = 1.95 (95% CI: 1.50–2.54), respectively (overall $P < 0.001$). Kaplan-Meier curves by cholesterol category are shown in Figure S2. The trend test confirmed a significant dose-response relationship between increasing cholesterol category and MI/CHD hazard ($P < 0.001$).

4. Discussion

This study demonstrated that both diabetes and total cholesterol are significant, independent predictors of MI/CHD risk in the Framingham cohort. Diabetic participants had substantially worse survival at every time point, with 43% experiencing MI/CHD by 20 years compared to 12% of non-diabetic participants. After adjustment for age, the diabetes hazard ratio was attenuated from 4.14 to 3.21, underscoring age as an important confounder. The absence of a statistically significant interaction ($P = 0.36$) suggests the cholesterol effect does not vary meaningfully by diabetes status.

The secondary categorical analysis reinforces a graded, dose-response relationship between cholesterol levels and MI/CHD

risk, consistent with established clinical thresholds. The video-only strategy — and by analogy, passive risk information alone — does not appear sufficient; combined intervention on both modifiable risk factors simultaneously is warranted.

4.1 Limitations

The Framingham cohort was recruited from a single Massachusetts city in 1948 and may not generalize to contemporary or more diverse populations. Diabetes and cholesterol were measured only at baseline, precluding assessment of time-varying exposure. The small number of diabetic participants ($N = 114$, 2.6%) limits precision for diabetes-stratified analyses, as evidenced by wide CIs and the inability to estimate the upper bound of median survival. Residual confounding from unmeasured factors (e.g., diet, medication use) cannot be excluded.

Author Contributions

Michelle Navarrete Vega and Chiamaka Lebechi contributed equally to the conceptualization, data analysis, interpretation of results, and manuscript preparation for this project.

References

- [1] National Heart, Lung, and Blood Institute. Framingham Heart Study. <https://www.nhlbi.nih.gov/science/framingham-heart-study-fhs>

Figures

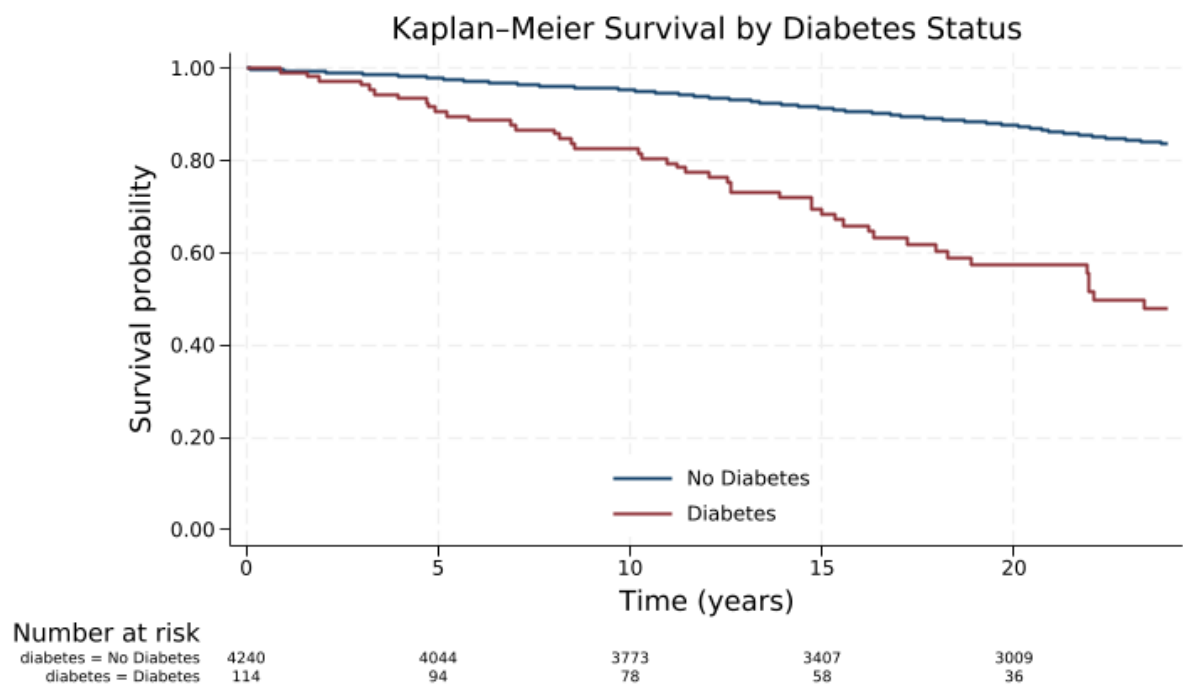
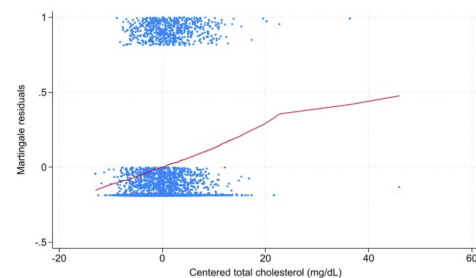


Figure 1. Kaplan-Meier survival curves for time to first myocardial infarction or coronary heart disease death, by diabetes status. Numbers at risk at 0, 5, 10, 15, and 20 years are shown below the figure. Legend is placed inside the figure.

Supplemental Materials

Panel A:



Panel B:

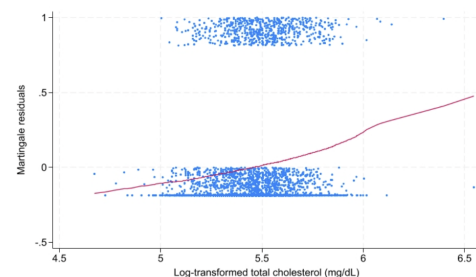


Figure S1. Martingale residual plots assessing linearity of total cholesterol in the Cox proportional hazards model. Panel A: centered total cholesterol (non-linear). Panel B: log-transformed, centered total cholesterol (approximately linear).

Table S1. Assessment of confounding for the association between diabetes, total cholesterol, and MI/CHD.

Model	HR (DM)	% Chg	HR (Chol)	% Chg
Base	3.92	—	1.07	—
+ Age	3.21	18.1%	1.06	0.9%
+ Sex	3.84	2.0%	1.08	0.9%
+ Education	3.71	5.4%	1.08	0.9%

≥10% change = evidence of confounding.

HR = hazard ratio; DM = diabetes; Chol = cholesterol.

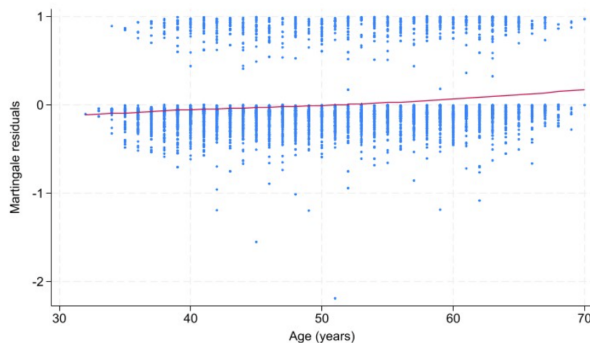
Table S2. Univariate and multivariable Cox proportional hazards models for risk of MI/CHD.

Variable	Univariate HR (95% CI)	Multivariable HR (95% CI)
Diabetes (Yes vs. No)	4.14 (3.05–5.63)*	3.21 (2.35–4.39)*
Total cholesterol [†]	1.07 (1.06–1.09)*	1.06 (1.04–1.07)*
Age (per year)	—	1.04 (1.03–1.06)*

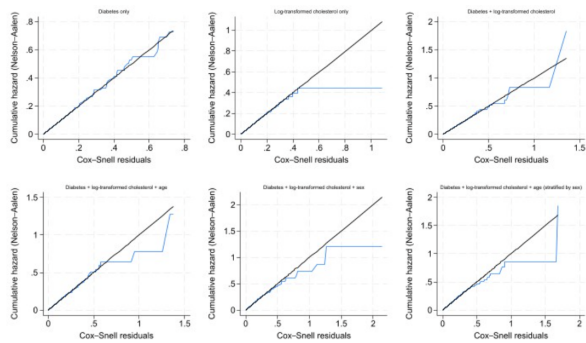
* $p < 0.001$. Multivariable model adjusted for age.

[†] Log-transformed, centered, scaled per 10 mg/dL.

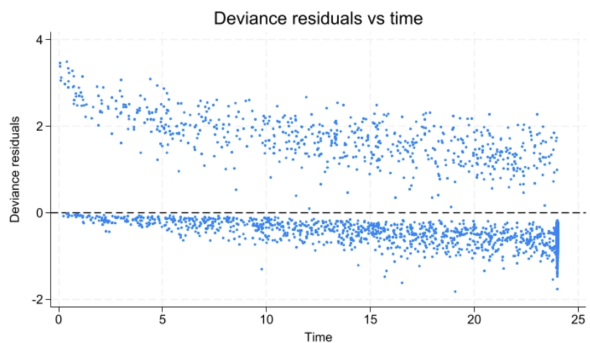
Panel A:



Panel B:



Panel C:



Panel D:

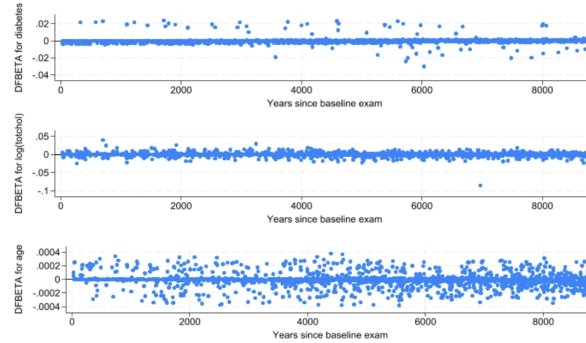


Figure S2. Martingale residual plots for age show no evidence of non-linearity (Panel A). Cox-Snell residual plots indicated that overall model fit improved with the addition of diabetes, log cholesterol, and age, stratified by sex (Panel B). Deviance residual plots showed that most observations were centered around zero (Panel C). DFBETA plots indicated that individual observations had minimal influence on model coefficients (Panel D).

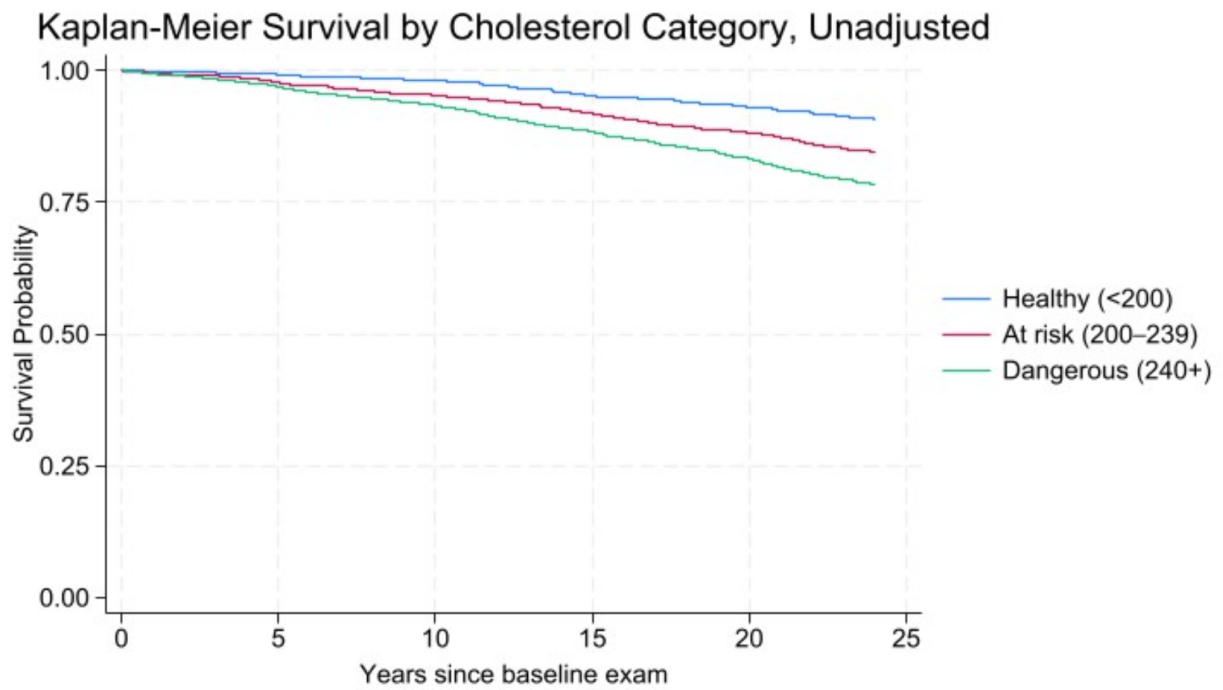


Figure S3. Kaplan-Meier survival curves for time to first myocardial infarction or coronary heart disease death, by total cholesterol category. Healthy (<200 mg/dL), at risk (200–239 mg/dL), dangerous (≥ 240 mg/dL). Log-rank $P < 0.001$.